Product Requirements Document

AI-Enabled Gift Recommendation System

Minimum Viable Product (MVP)

Version 1.0 | April 2026

1. Context

Our company is a fast-fashion retail leader whose mobile app serves as the primary commerce and social platform for U.S. teenage shoppers. Over the past three years, revenue has grown 10x, driven predominantly by customers purchasing for themselves. However, growth has decelerated as the core U.S. teenage demographic approaches saturation.

The app is already highly social: customers browse together, share looks, and show off purchases—mirroring the "mall with friends" experience online. Critically, most customers are already connected to 2–3 close friends or family members within the platform, and the company holds rich historical data on browsing, purchasing, and demographics.

Several expansion avenues (new demographics, new product categories, international expansion, store-branded goods) were evaluated but rejected due to cost, complexity, or timeline constraints. The strongest near-term opportunity is to capitalize on the social graph already embedded in the app by encouraging gift purchasing—a behavior that is natural for the demographic and occasion-driven.

2. MVP Goal

Launch an AI-powered Gift Recommendation System within the existing mobile app by the upcoming holiday season. The system will surface personalized gift suggestions to shoppers for their in-app connections—friends and family—leveraging existing purchase history, social data, and demographic signals.

This MVP will:

- Increase total Gross Merchandise Value (GMV) by converting social browsing sessions into gift purchases.
- Expand the average order size by introducing a new gifting purchase occasion.
- Deepen platform engagement by making the app the go-to destination for holiday and occasion-based gifting.
- Generate gifting-specific behavioral data to inform future iterations.

3. Target Customer Segment

For MVP, the target segment is existing U.S. teenage customers (the current core user base). Key characteristics:

Attribute	Detail
Geography	United States (domestic only for MVP)
Age	Teenagers (primary existing demographic)
Brand Affinity	NPS of 75 — highly loyal and brand-positive
Purchase Behavior	35% repeat purchasers (2+ purchases in past 12 months)
Social Connections	Each customer connected to ~2–3 friends/family in-app
Commerce Readiness	Address and payment already on file — low friction
Data Available	Browse history, purchase history, demographic data

4. Timeframe

The MVP must be live within 6 months to capture the upcoming holiday gifting season, which represents the highest-value gifting opportunity of the year.

Phase	Scope	Timeframe
Phase 1: Foundation	Data pipeline, model training, gift tagging flow	Months 1–2
Phase 2: In-App MVP	Recommendation UI, recipient selection, checkout	Months 3–4
Phase 3: Forecasting & Comms	Email triggers, stakeholder dashboard	Month 5
Phase 4: QA & Launch	Testing, performance tuning, holiday launch	Month 6

5. Define Success Metrics

The following KPIs and metrics will be used to evaluate MVP success and guide future iterations. Each metric has a clear data source and a defined measurement approach.

5.1 Gifting Frequency

Measures how often customers are initiating and completing gift purchases, indicating whether the feature is being discovered and used repeatedly.

Metric	Definition	Data Source
Gift Order Rate	% of all orders flagged as gift purchases	Checkout gift flag
Repeat Gifting Rate	% of gifting users who make 2+ gift purchases in 90 days	Order history, gift flag
Recommendation Click-Through Rate (CTR)	% of gift recommendation impressions that result in a product view	App event tracking
Gift Recommendation Conversion Rate	% of gift CTRs that lead to a completed purchase	App event + order data

5.2 Gifting Value

Measures the monetary contribution of gift purchases, capturing whether gifting drives incremental revenue and premium basket sizes.

Metric	Definition	Data Source
Average Gift Order Value (AGOV)	Mean order value for orders tagged as gifts	Order data + gift flag
Gift Revenue as % of Total Revenue	GMV from gift orders divided by total GMV	Order + revenue data
Incremental Revenue per Gifting User	Additional spend by users who gift vs. non-gifters (same cohort)	Order history, cohort analysis

5.3 Overall Marketplace Value

Measures the broader business impact of the gifting feature on platform health, customer retention, and total GMV growth.

Metric	Definition	Data Source
Total GMV Growth Rate (YoY)	Year-over-year change in GMV for holiday period	Revenue reporting
New Gifting-Driven Customer Acquisitions	# of first-time buyers who entered via receiving a gift	Account creation + order data
Customer Retention Rate (Gifters vs. Non-Gifters)	12-month repurchase rate, segmented by gifting behavior	Order history, cohort

6. Select AI Models and Data

6.1 AI-Powered Recommendation System

Selected Model: Hybrid Recommendation System

A Hybrid Recommendation System combines Collaborative Filtering ("shoppers like this person also purchased...") with Content-Based Filtering ("based on products this recipient has shown interest in..."). This is the most appropriate model because:

- The platform already has rich behavioral data (browse, purchase, social) that is ideal for collaborative filtering.
- Cold-start problem for new gift recipients (who may have limited data) is mitigated by content-based signals from their social profile and demographic data.
- Gift recommendations require both social context (who is the recipient?) and product context (what would they like?) — a hybrid approach handles both naturally.

Data Source	Type	How It Enhances the Model
Shopper's own purchase history (self)	Exogenous	Reveals brand preferences and style affinity to seed collaborative signals
Past gift purchases for others	Exogenous	Identifies gift-giving patterns unique to this shopper (e.g., preferred price range, categories)
Most popular products sent as gifts (platform-wide)	Exogenous	Provides crowd-sourced signals for collaborative filtering, especially for cold-start recipients
Recipient social connections & communications	Exogenous	Infers recipient's interests from in-app activity and social posts
Gift occasion (from checkout question)	Exogenous	Enables occasion-specific filtering (birthday vs. holiday vs. general); supports recurring occasion reminders
Recipient birthdate / horoscope sign	Exogenous	Enables proactive birthday gift recommendations; horoscope adds a low-friction, fun alternative to explicit age collection
Recipient demographic & location (zip code)	Exogenous	Surfaces locally trending products to increase relevance of recommendations
Personal interests (inferred from browsing/social)	Exogenous	Feeds content-based filtering to match recipient tastes (hobbies, music, movies, memes)

6.2 AI-Powered Forecasting Tool

Selected Model: Time Series Forecasting (Autoregressive + Exogenous Inputs)

A time series forecasting model will predict gift order volume and GMV by week/day, enabling the business to plan inventory, staffing, and marketing campaigns ahead of peak gifting periods (holidays, Valentine's Day, etc.). An ARIMA or Prophet model augmented with exogenous inputs is recommended for its interpretability and suitability with retail seasonality patterns.

Data Source	Type	How It Enhances the Model
Historical gift order volume (weekly/daily)	Autoregressive	Core time-series signal from which the model learns seasonal trends and baseline demand
Historical GMV by order type (gift vs. self-purchase)	Autoregressive	Enables revenue-level forecasting and helps isolate gifting's incremental contribution
U.S. holiday calendar (Thanksgiving, Christmas, Valentine's Day, etc.)	Exogenous	Major driver of gifting demand spikes; allows model to anticipate and quantify seasonal peaks
Email / push campaign send dates and open rates	Exogenous	Captures marketing-driven demand lifts to distinguish organic vs. campaign-generated gifting activity
Recipient birthdate data (aggregated monthly)	Exogenous	Creates predictable micro-peaks as customers receive birthday gift reminders at scale
New social connection formation rate (weekly)	Exogenous	Growing social graph expands addressable gifting occasions; correlates with future gifting demand

7. Write User Stories

The following user stories define the capabilities required for the MVP. They are organized by functional area.

7.1 Gift Recommendation — In-App

User Story 1

As a teenage shopper browsing the app, I want to see AI-generated gift recommendations for my in-app friends and family when I open the app near a known gifting occasion (e.g., a friend's upcoming birthday, the holiday season) so that I can quickly find a great gift without leaving the app, making the experience feel thoughtful and effortless.

Acceptance Criteria:

- Recommendations appear on a dedicated 'Gift Ideas' surface in the app for occasions within 30 days.
- Each recommendation card shows product image, name, price, and a 'Send as Gift' CTA.
- At least 5 personalized product recommendations are generated per recipient.
- Recommendations are generated using the Hybrid Recommendation model fed by the data sources defined in Section 6.1.

User Story 2

As a shopper who has found a gift I want to send, I want to add a gift message, choose gift wrapping, and send the item directly to my friend's address stored in the app (or a new address) in a single checkout flow so that I can complete a gift purchase in under 2 minutes without friction or confusion.

Acceptance Criteria:

- A 'Gift this item' option is available on any product detail page.
- Shopper can select a recipient from their in-app social connections.
- Gift message field (max 150 characters) is presented before checkout.
- Order is flagged as a gift purchase in the system for data tracking purposes.

7.2 Gift Recommendation — Email

User Story 3

As a shopper with friends or family in the app whose birthday is approaching, I want to receive a personalized email 7 days before each friend's birthday, featuring AI-curated gift ideas based on that friend's interests and my past purchase behavior so that I don't miss important gifting moments and I feel like the app understands my relationships.

Acceptance Criteria:

- Email is triggered automatically 7 days before a connected friend's/family member's birthday (where birthdate data is available).
- Email includes 3 personalized product recommendations with images, names, prices, and a direct 'Shop this Gift' CTA linking to the gift checkout flow.
- Email is mobile-optimized and consistent with brand standards.
- Unsubscribe from birthday gift email reminders is available in one click.

7.3 Forecasting Dashboard — Internal Stakeholders

User Story 4

As a Product Manager or Business Analyst, I want to access a web-based dashboard that displays forecasted gift order volume and gift GMV by week for the next 8 weeks, alongside actuals, key KPI trends, and model confidence intervals so that I can proactively identify underperformance, plan inventory, and make data-informed decisions about marketing campaigns and feature iterations.

Acceptance Criteria:

- Dashboard is accessible via web browser (no app install required) with role-based access control.
- Charts include: (1) Forecasted vs. actual gift orders by week, (2) Gift GMV vs. total GMV trend, (3) Top recommended products by CTR and conversion, (4) Gifting frequency cohort breakdown.

- Summary statistics visible at a glance: AGOV, Gift Order Rate, Repeat Gifting Rate, Recommendation CTR.
- Dashboard refreshes daily with prior-day actuals.

User Story 5

As a senior business stakeholder (VP, C-suite), I want to receive a weekly automated email every Monday morning containing a concise performance summary of the gift recommendation feature, including actual vs. forecasted GMV, key KPIs, and any anomalies flagged by the model so that I can stay informed on gifting performance without logging into a dashboard, and can quickly identify whether the initiative is on track to meet our holiday season targets.

Acceptance Criteria:

- Email is sent every Monday by 8:00 AM local time.
- Email includes: actual gift GMV for the prior week, forecast for the upcoming week, week-over-week change in Gift Order Rate and AGOV, and a one-sentence model commentary on notable trends.
- Email links to the full stakeholder dashboard for deeper analysis.

7.4 Backlog: Gift Purchase Identification

User Story 6

As a Data Scientist or AI Engineer, I want to have every order in the system explicitly tagged as a 'gift purchase' or 'self-purchase' via a checkout flow question presented to all shoppers so that the recommendation and forecasting models have high-quality, labeled ground truth data to train on — improving accuracy over time without relying on inference alone.

Acceptance Criteria:

- A single-question prompt ('Is this a gift for someone else?') is displayed during checkout, before the order confirmation step.
- If the shopper selects 'Yes,' they are optionally prompted to select or confirm the recipient from their social connections, and enter the gift occasion (birthday, holiday, just because, etc.).
- The gift flag and occasion data are stored at the order level and made available to the recommendation model training pipeline.
- The checkout question is optional (not blocking) to avoid increasing cart abandonment rate.

8. Stakeholder Considerations

Stakeholder	Key Input Required / Dependencies
Engineering Lead	Estimate effort for recommendation API, gift checkout flow, dashboard infrastructure, and email trigger system. Flag dependencies on data pipeline readiness.

Stakeholder	Key Input Required / Dependencies
Data Scientist	Define training data schema, labeling standards for gift flag, model evaluation metrics, and retraining cadence.
Design Lead	Design 'Gift Ideas' in-app surface, gift checkout flow, email templates, and dashboard UI. Collaborate with Engineering on feasibility trade-offs.
AI Product Manager	Own this PRD; prioritize backlog; track KPIs post-launch; define iteration roadmap based on holiday season performance.

— End of Document —